

Mathematical Models – Introduction

From a differential equation to a transfer function.

Ogata, *Modern Control Engineering*, Ch. 2.

In this tutorial you will:

- recall the Laplace transform pairs that make LTI analysis algebraic,
- turn a governing ODE into a transfer function $G(s) = Y(s)/U(s)$, and
- read poles and DC gain off $G(s)$ with `pole` and `dcgain`.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A mathematical model of a dynamic system is the set of equations that represent its dynamics. For linear, time-invariant (LTI) systems governed by a constant-coefficient linear ODE

$$a_n \frac{d^n y}{dt^n} + \dots + a_1 \dot{y} + a_0 y = b_m \frac{d^m u}{dt^m} + \dots + b_1 \dot{u} + b_0 u$$

the Laplace transform converts the ODE (with zero initial conditions) into an algebraic relationship between the transforms of the input and output, $U(s)$ and $Y(s)$.

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The Laplace Transform

The Laplace transform of a time function $f(t)$, $t \geq 0$, is

$$F(s) = \mathbf{L}[f(t)] = \int_0^\infty f(t)e^{-st} dt$$

Two properties make it the natural tool for LTI systems:

- Differentiation: $\mathbf{L}[\dot{f}(t)] = sF(s) - f(0)$
- Linearity: $\mathbf{L}[af(t) + bg(t)] = aF(s) + bG(s)$

Common transform pairs used throughout this repository:

$$\mathbf{L}[1(t)] = \frac{1}{s}, \quad \mathbf{L}[t] = \frac{1}{s^2}, \quad \mathbf{L}[e^{-at}] = \frac{1}{s+a}, \quad \mathbf{L}[\sin(\omega t)] = \frac{\omega}{s^2 + \omega^2}$$

```
syms t s
f1 = heaviside(t);      % unit step
f2 = t;                 % unit ramp
f3 = exp(-2*t);         % decaying exponential
f4 = sin(3*t);          % sinusoid

F1 = laplace(f1,t,s);
F2 = laplace(f2,t,s);
```

```
F3 = laplace(f3,t,s);
F4 = laplace(f4,t,s);

disp('Step, ramp, exponential, sinusoid transforms:')
disp([F1 F2 F3 F4])
```

```
Step, ramp, exponential, sinusoid transforms:
[1/s, 1/s^2, 1/(s + 2), 3/(s^2 + 9)]
```

From ODE to Transfer Function

With zero initial conditions, taking the Laplace transform of the governing ODE above and solving for the ratio $Y(s)/U(s)$ gives the **transfer function**

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_ms^m + \dots + b_1s + b_0}{a_ns^n + \dots + a_1s + a_0}$$

As a first example, consider

$$\ddot{y} + 3\dot{y} + 2y = u$$

Taking the Laplace transform term by term: $s^2Y(s) + 3sY(s) + 2Y(s) = U(s)$, so

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

```
G = tf(1,[1 3 2]);
G
```

```
G =
```

$$\frac{1}{s^2 + 3s + 2}$$

```
Continuous-time transfer function.
```

The poles of $G(s)$ are the roots of the denominator polynomial; they determine the natural (unforced) response of the system.

```
poles_G = pole(G)
```

```
poles_G =
```

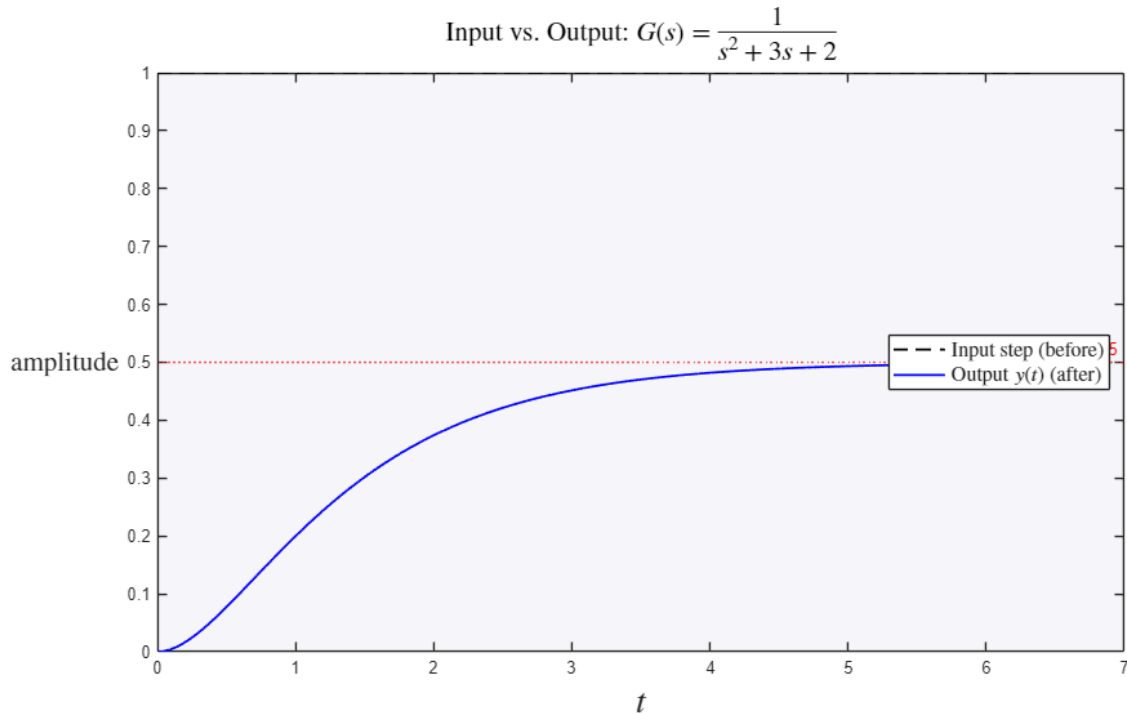
```
-2
-1
```

A unit-step response confirms the system settles to its DC gain, $G(0) = 1/2$, consistent with the final value theorem $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sG(s) \frac{1}{s} = G(0)$.

```

figure
[ys,ts] = step(G);
plot(ts, ones(size(ts)), 'k--', ts, ys, 'b', 'LineWidth', 1.3)
yline(0.5, 'r:', 'DC gain 0.5')
legend('Input step (before)', 'Output $y(t)$ (after)', 'Interpreter', 'latex', 'FontSize', 12, 'Location', 'east')
title('Input vs. Output: $G(s)=\frac{1}{s^2+3s+2}$', 'Interpreter', 'latex', 'FontSize', 16)
ylabel('amplitude', 'Interpreter', 'latex', 'FontSize', 16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)

```



The remaining files in this directory build up the physical systems (mechanical, electrical, fluid, thermal) whose governing equations produce transfer functions of this type, plus the block-diagram algebra used to combine them into full control systems.

Try it yourself

- Swap $G = \text{tf}(1, [1 \ 3 \ 2])$ for one with complex poles, e.g. $\text{tf}(1, [1 \ 1 \ 4])$, and notice the step response now overshoots and rings.
- Use $\text{dcgain}(G)$ to predict the final value before you plot it.